IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

Corrected copy

SMART GRID TECHNOLOGIES

Wednesday, 27 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks

Use answer book A to answer Question 1, answer book B to answer questions 2&3 and answer book C to answer question 4.

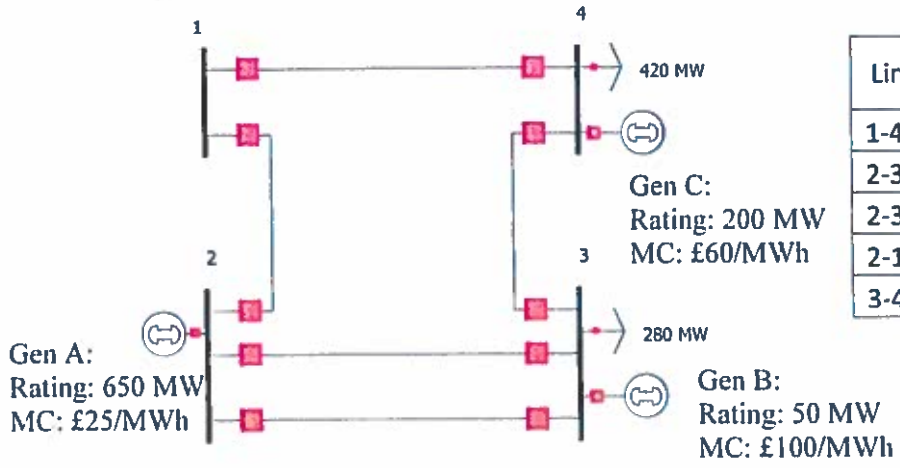
Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.H. Tindemans, G. Strbac, S.H. Tindemans

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The Questions

1. An industrial complex runs an isolated 4-node system as shown in Figure 1. The system is connected with 2 customer loads and 3 customer generators (i.e. Gen A, B and C) with different ratings and marginal fuel costs. Loads are connected at bus 3 (280 MW) and bus 4 (420 MW). The rating and impedance of each circuit is listed in the table below.



Line	X (p.u.)	Capacity (MW)
1-4	0.100	420
2-3	0.175	350
2-3	0.175	350
2-1	0.250	420
3-4	0.350	500

Figure 1: A 4-node system

a) Determine the economic dispatch, which minimizes the operating cost. [2]

b) Calculate the flows at each line. [3]

c) If the system should withstand a single unplanned line outage, carry out contingency analysis and determine the critical contingency. [4]

d) In order to operate securely in preventive mode, determine how generation output should be re-dispatched. [4]

e) Calculate the cost of security. [2]

f) In order to reduce the operating cost, the system operator decides to install a Quadrature-booster. Identify the optimal location for installing the Quadrature-booster and its setting (phase angle). [5]

2. Quantification of distribution network reliability

a) Provide the definitions for CML and CI. What are typical CML and CI values for UK distribution networks?

[3]

b) Consider a section of a distribution network that serves 3000 customers. Table 2.1 lists the outage events that occurred on this network in 2015.

Customers affected	Duration
200	3 hours
500	10 minutes
1500	2 minutes
100	2 hours
20	2 days
300	30 minutes
1000	1 hour

Table 2.1: outage events

Compute the CML and CI values for this section of the network.

[3]

c) Consider the distribution network in Figure 2.1.

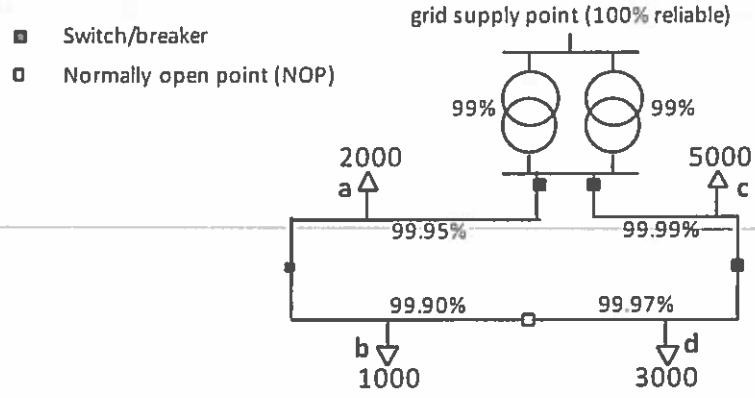


Figure 2.1: network diagram

The load points (a-d) are labelled with the number of customers connected to each load point. The maximum diversified load is 1kW per customer, with a 0.85 load factor. The capacity of each transformer is 15MVA. The availability of each component (transformer, feeder section) is listed alongside as a percentage. Component failures are assumed to be independent and to last for more than 3 minutes each. The network is radially operated, but the normally open point (NOP) provides a degree of redundancy: the NOP is normally open, but it can be closed to resupply (b) or (d) in case of an

upstream fault in the feeders at (a) or (c). You may assume that the normally open point reroutes power within 3 minutes.

The customers are a mixture of residential and SME customers, specified in Table 2.2. The customer interruption costs are £10,000/MWh for residential customers and £35,000/MWh for SMEs, and their average power consumption during outages is 0.4 kW (residential customer) and 2.5 kW (SME customer).

Load point	Residential customers	SME customers
a	1900	100
b	1000	0
c	4700	300
d	2700	300

Table 2.2: customer classification

i) Derive the expressions for the probabilities that each of the the load points is connected to the grid supply point (one expression for each load point). Then compute the expected CML value for this network.

[7]

ii) Compute the expected annual cost of interruptions under these assumptions, using the load point supply probabilities computed in (i).

[4]

d) Explain the Willingness-to-Pay (WTP) and Willingness-to-Accept (WTA) approaches to evaluating customer interruption costs. Do you expect similar values when using WTA and WTP? Explain which approach you consider most suitable as a basis for interruption costs models.

[3]

3. Reliability models and Monte Carlo analysis

a) The reliability of a 5km underground cable is modelled as a repairable two-state system. Its failure rate is 0.023/km/year and its availability is 0.9998. What is the mean time to repair?

[3]

b) Consider an automated monitoring and control system that has two failure modes: (1) a 'soft' failure that is automatically detected and rectified by reinitialising the system; (2) a 'hard' failure that requires a field visit by an engineer. We will model this process as a Markov chain with 3 states and the following transition rates: soft failure rate: 1in 20 days; hard failure rate: 1 in 230 days; recovery rate from soft failure: 0.5 per hour; hard failure repair rate: 0.5 per day.

i) Draw the corresponding continuous-time Markov chain model, using consistent units (hours or days) for all transition rates.

[2]

ii) Derive and draw a discrete time Markov chain for the process in (b), assuming a time step of 1 hour. Include all non-zero transition probabilities. Discuss whether the discrete time Markov chain with a time step of 1 hour is a good approximation of the continuous time process.

[3]

c) Consider a demand aggregator that controls the load of a portfolio of customers (large electricity consumers). It uses this load control ability to offer a load shedding service to National Grid. When activated, an incoming load shedding signal from the National Grid control room is processed by two redundant computer systems. The computers then select appropriate loads to shed and send the necessary instructions over separate communication links. There is one communication link per customer, but this link is shared between both computer systems. Assume that the load shedding signal is sent to *two* customers.

To model the reliability of this system we will assume that the National Grid load shedding signal is 100% dependable. All other components (computer system, communications links, load shedding equipment at customer premises) are modelled as repairable systems. Failures are assumed to be statistically independent.

i) Draw a reliability block diagram for the successful load shedding of *both* customers.

[2]

ii) Draw a reliability block diagram for the successful load shedding of *at least one* of the two customers.

[2]

iii) Assuming an availability of 98% for each component, quantify the probability of success for each of the diagrams in (i) and (ii).

[2]

d) Write down the formula for the *Monte Carlo estimator* for the risk $R = E[X]$, using N samples of the random variable X.

[2]

e) A snapshot Monte Carlo simulation is used to estimate the expected cost of interruptions. After N=10,000 samples the best estimate of the cost is £108,762, with a standard error of £2581.

i) How many samples will be required (approximately, in total) to reduce the *standard error* of the mean to £1000?

[2]

ii) How many samples will be required (approximately) to reduce the width of the *95% confidence interval* for the mean to £1000?

[2]

4. Flexible demand and big data

a) Explain the principles of centralised and price-based scheduling of flexible demand appliances. What are the advantages and disadvantages of each of these approaches?

[4]

b) A simple power system, operating at a two-hour horizon, includes:

- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 50s_t^2$ (£) and a maximum output limit $s^{max} = 10\text{MW}$
- inflexible demand appliances, consuming power $D_1 = 3\text{MW}$ (at hour 1) and $D_2 = 2\text{MW}$ (at hour 2)
- 1000 identical flexible demand appliances with continuously adjustable power d_t (kW), scheduling interval including hours 1 and 2, total energy required $E = 5\text{kWh}$ and maximum power limit $d^{max} = 5\text{kW}$

i) Under centralised scheduling, what is the optimal power demand of each of flexible demand appliances at each of the two hours? What is the total generation cost?

[2]

ii) Assume that price-based scheduling is employed. What should be the relation between the prices at hour 1 and 2 (denoted by λ_1 and λ_2 respectively)? What is the resulting power demand of each of the flexible demand appliances? What is the total generation cost? State whether it is higher or lower than the total generation cost calculated in (i), and explain the reasons for this outcome.

[4]

c) List the primary sources of big data in power systems.

[2]

d) Briefly outline three big data analysis techniques.

[2]

e) Big data analysis can be used to: monitor asset health, characterise residential electricity consumption and improve large-scale security assessment platforms. Describe how each of these three tasks has been carried out in the past and give an example of how each task can be improved by using big data analysis.

[6]